

MATH 1210 Worksheet 5 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

Remark: The square of a matrix A is defined by $A^2 = A \cdot A$. Similarly, we have $A^3 = A \cdot A^2 = A \cdot A \cdot A$, and $A^4 = A \cdot A^3 = A \cdot A \cdot A \cdot A$. Recursively we define $A^n = A \cdot A^{n-1}$.

1. Show that

$$\begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^3$$

is the zero matrix.

Solution: First we compute

$$\begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence

$$\begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 3 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

2. Prove by induction that

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}$$

for any $n \geq 1$.

Solution: Let us denote by $P(n)$ the statement

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^1 = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$ and equal to the right hand side. $\begin{pmatrix} 2 \cdot 1 + 1 & -4 \cdot 1 \\ 1 & 1 - 2 \cdot 1 \end{pmatrix} = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$. Hence $P(n)$ is true for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^k = \begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix}. \quad (1)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^{k+1} = \begin{pmatrix} 2k+3 & -4k-4 \\ k+1 & -1-2k \end{pmatrix}. \quad (2)$$

Using the induction hypothesis (1), the left-hand side of (2) is

$$\begin{aligned} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^{k+1} &= \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^k \\ &= \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} \\ &= \begin{pmatrix} 3(2k+1) - 4k & 3(-4k) - 4(1-2k) \\ (2k+1) - k & -4k - (1-2k) \end{pmatrix} \\ &= \begin{pmatrix} 2k+3 & -4k-4 \\ k+1 & -1-2k \end{pmatrix} \end{aligned}$$

Hence, the proposition $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

3. Consider the vectors $\mathbf{u} = \langle 1, -2, -3 \rangle$, $\mathbf{v} = \langle 2, 2, -1 \rangle$ and $\mathbf{w} = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, the points $P(2, 3, -1)$ and $Q(1, 0, 4)$.

- (a) Is the vector joining the points P and Q perpendicular to $\mathbf{u} + \mathbf{v}$?

Solution: We have $\mathbf{u} + \mathbf{v} = \langle 1, -2, -3 \rangle + \langle 2, 2, -1 \rangle = \langle 3, 0, -4 \rangle$. The vector joining the points P and Q is $\mathbf{PQ} = \langle -1, -3, 5 \rangle$. The dot product

$$\mathbf{PQ} \cdot (\mathbf{u} + \mathbf{v}) = \langle -1, -3, 5 \rangle \cdot \langle 3, 0, -4 \rangle = (-1) \cdot 3 + 5 \cdot (-4) = -3 - 20 = -23$$

is nonzero, hence the vectors are not perpendicular.

- (b) Find a unit vector parallel to $2\mathbf{u} - \mathbf{w}$.

Solution: The components of $\mathbf{w}$ are $\mathbf{w} = \langle 3, -4, 2 \rangle$.

$$\begin{aligned} 2\mathbf{u} - \mathbf{w} &= 2\langle 1, -2, -3 \rangle - \langle 3, -4, 2 \rangle \\ &= \langle 2, -4, -6 \rangle - \langle 3, -4, 2 \rangle \\ &= \langle -1, 0, -8 \rangle. \end{aligned}$$

The length of $2\mathbf{u} - \mathbf{w}$ is

$$|2\mathbf{u} - \mathbf{w}| = \sqrt{(-1)^2 + (-8)^2} = \sqrt{65}.$$

The vector

$$\frac{2\mathbf{u} - \mathbf{w}}{|2\mathbf{u} - \mathbf{w}|} = \left\langle -\frac{1}{\sqrt{65}}, 0, -\frac{8}{\sqrt{65}} \right\rangle$$

is now a unit vector, and parallel to $2\mathbf{u} - \mathbf{w}$.

- (c) Find a vector of length 2 in the opposite direction to $\mathbf{u}$

Solution: $|\mathbf{u}| = \sqrt{1^2 + (-2)^2 + (-3)^2} = \sqrt{14}$. So

$$-2 \frac{\mathbf{u}}{|\mathbf{u}|} = \left\langle \frac{-2}{\sqrt{14}}, \frac{4}{\sqrt{14}}, \frac{6}{\sqrt{14}} \right\rangle$$

- (d) Find a vector of length 3 perpendicular to $\mathbf{v}$ and to the x -axis.

Solution: Cross product is

$$\mathbf{v} \times \langle 1, 0, 0 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -1 \\ 0 & 0 & 1 \end{vmatrix} = \mathbf{i} \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{j} = \langle 2, -2, 0 \rangle$$

Has length $|\mathbf{v} \times \langle 1, 0, 0 \rangle| = \sqrt{2^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$ so

$$3 \frac{\mathbf{v} \times \langle 1, 0, 0 \rangle}{|\mathbf{v} \times \langle 1, 0, 0 \rangle|} = \left\langle \frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}, 0 \right\rangle$$

4. Consider the vectors $\mathbf{u} = \langle m, 2m, -1 \rangle$ and $\mathbf{v} = \langle 2m, 1, 3 \rangle$ and $\mathbf{w} = \langle 1, 0, m \rangle$ where m is a real number.

- (a) For which values of m are the vectors $\mathbf{u}$ and $\mathbf{v}$ perpendicular?

Solution: The vectors are perpendicular if and only if the dot product $\mathbf{u} \cdot \mathbf{v} = 2m^2 - 2m - 3$ is zero. So we need to find m that is a solution of the quadratic equation $2m^2 - 2m - 3 = 0$. The discriminant of the equation is $(-2)^2 - 4 \cdot 2 \cdot (-3) = 28$. The two roots of the quadratic equation are

$$\frac{2 \pm \sqrt{28}}{4} = \frac{1 \pm \sqrt{7}}{2}.$$

Hence $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if $m = \frac{1+\sqrt{7}}{2}$ or $m = \frac{1-\sqrt{7}}{2}$

- (b) For which values of m are the vectors $\mathbf{u}$ and $\mathbf{w}$ perpendicular?

Solution: The dot product $\mathbf{u} \cdot \mathbf{w} = m - m = 0$ is zero, so the vectors are perpendicular for any number m .

5. Find an equation of the plane containing the points $P_1(2, 3, 4)$, $P_2(1, -1, 0)$ and $P_3(3, 1, 1)$.

Solution: The plane contains the segment joining P_1 to P_2 and the segment joining P_1 to P_3 . So the plane is parallel to the vectors $\mathbf{P}_1\mathbf{P}_2 = \langle -1, -4, -4 \rangle$ and $\mathbf{P}_1\mathbf{P}_3 = \langle 1, -2, -3 \rangle$. Hence, the cross product

$$\mathbf{P}_1\mathbf{P}_2 \times \mathbf{P}_1\mathbf{P}_3 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -4 & -4 \\ 1 & -2 & -3 \end{vmatrix} = \mathbf{i} \begin{vmatrix} -4 & -4 \\ -2 & -3 \end{vmatrix} - \mathbf{j} \begin{vmatrix} -1 & -4 \\ 1 & -3 \end{vmatrix} + \mathbf{k} \begin{vmatrix} -1 & -4 \\ 1 & -2 \end{vmatrix} = 4\mathbf{i} - 7\mathbf{j} + 6\mathbf{k} = \langle 4, -7, 6 \rangle$$

is a normal vector of the plane passing through the three points. We pick the point $P_1(2, 3, 4)$ and we get the equation

$$4(x - 2) - 7(y - 3) + 6(z - 4) = 0,$$

which can also be written

$$4x - 7y + 6z = 11.$$

6. Consider the plane $x + y - 4z = -3$.

(a) Which one of the following three points lies on the plane?

$$P_1(3, 2, 2) \quad P_2(5, 4, 3) \quad P_3(2, -1, 7)$$

Solution: To check if a point with coordinates lies on the plane, we have to verify if it satisfies the equation. We verify that P_1 satisfies the equation

$$x + y - 4z = 3 + 2 - 4 \cdot 2 = 5 - 8 = -3.$$

Hence P_1 lies on the plane. For the point P_2 we have

$$x + y - 4z = 5 + 4 - 4 \cdot 3 = 9 - 12 = -3,$$

hence P_2 is also on the plane. For the point P_3 we have

$$x + y - 4z = 2 + (-1) - 4 \cdot 7 = 1 - 28 = -27,$$

hence P_3 does not lie on the plane.

(b) Is the line

$$\begin{cases} x = 2 + 3t \\ y = -1 + 5t \\ z = 1 + 2t \end{cases}$$

contained in the plane?

Solution: A point $Q(x, y, z)$ lies on the line if it satisfies the parametric equation, that is $x = 2 + 3t$, $y = -1 + 5t$ and $z = 1 + 2t$ for some real number t . To verify if any point on the line also lies on the plane, we must verify that any such point also satisfies the equation of the plane $x + y - 4z = -3$. We verify by substituting the coordinates of Q into the equation:

$$x + y - 4z = (2 + 3t) + (-1 + 5t) - 4 \cdot (1 + 2t) = 2 + 3t - 1 + 5t - 4 - 8t = -3.$$

This implies that any point Q on the line satisfies the equation of the plane, hence lies on the line. It follows that the line is contained in the plane.

7. Find an equation parallel to the xz -plane and containing the point $P(2, 2, 5)$.

Solution: A normal vector to the xz -plane is the vector $\langle 0, 1, 0 \rangle$. Hence an equation of the plane parallel to the xz -plane and passing through P is $y - 2 = 0$.

8. Find an equation of the plane through the point $P(1, -1, 2)$ and intersecting the plane $2x - 3y + z = 5$ along the line

$$\begin{cases} x = 1 + t \\ y = 2 + t \\ z = 9 + t \end{cases}$$

Solution: The vector $\mathbf{v} = \langle 1, 1, 1 \rangle$ is parallel to the line. Moreover, setting $t = 0$ we find that $Q(1, 2, 9)$ is a point on the line. Hence, the plane that we are looking for is parallel to the two vectors $\mathbf{v}$ and $\mathbf{PQ} = \langle 0, 3, 7 \rangle$. Hence, a normal vector of the plane is

$$\mathbf{v} \times \mathbf{PQ} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & 3 & 7 \end{vmatrix} = \mathbf{i} \begin{vmatrix} 1 & 1 \\ 3 & 7 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & 1 \\ 0 & 7 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = 4\mathbf{i} - 7\mathbf{j} + 3\mathbf{k} = \langle 4, -7, 3 \rangle.$$

So an equation of the plane through P and the given line is

$$4(x - 1) - 7(y + 1) + 3(z - 2) = 0,$$

or equivalently

$$4x - 7y + 3z = 17.$$

9. Find symmetric equations of the line

$$\begin{cases} x = 2 + 3t \\ y = 3 - 2t \\ z = 1 + 2t \end{cases}$$

Then find two planes that intersect along the line.

Solution: The symmetric equations are given by

$$\frac{x - 2}{3} = -\frac{y - 3}{2} = \frac{z - 1}{2}.$$

Hence the two planes

$$2(x - 2) + 3(y - 3) = 0$$

$$2(y - 3) - 2(z - 1) = 0$$

intersect along. The equations of the two planes can be simplified to

$$2x + 3y = 13$$

$$2y - 2z = 4$$

10. For which values of p and q is the following matrix in reduced row echelon form?

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 - q & 1 \\ 0 & 0 & p^2 + qp - q - 1 \end{pmatrix}$$

Solution: First we note that the lower right entry cannot be a leading 1 since there is a 1 above it. Hence it must be 0 and the values p and q must solve the equation

$$p^2 - qp - q - 1 = 0. \quad (3)$$

The second entry on the diagonal can be either 0 or 1. Let us consider the two cases separately:

- Case 1: suppose that the second entry on the diagonal is $3 - q = 0$. Hence $q = 3$ and the corresponding values of p are given by solving equation (3) for $q = 3$:

$$p^2 - 3p - 4 = 0.$$

This equation has two solutions $p = \frac{1 \pm \sqrt{25}}{2} = \frac{1 \pm 5}{2}$. Thus if $q = 3$ then $p = 3$ or $p = -2$.

- Case 2: suppose that the second entry on the diagonal is $3 - q = 1$. Hence $q = 2$ and the corresponding values of p are given by solving equation (3) for $q = 0$:

$$p^2 - 2p - 3 = 0.$$

This equation has two solutions $p = \frac{2 \pm \sqrt{16}}{2} = 1 \pm 2$.

In conclusion we get the following 4 possible values for (p, q) :

$$(3, 3), (-2, 3), (-1, 2), (3, 2).$$

11. What is the intersection of the planes $-2x + 5y + 3z = 5$ and $-2y + 3z = -1$? Describe it geometrically.

Solution: The intersection points are exactly the solutions of the system of linear equations

$$\begin{cases} -2x + 5y + 3z = 5 \\ -2y + 3z = -1. \end{cases}$$

The augmented matrix of the system

$$\left(\begin{array}{ccc|c} -2 & 5 & 3 & 5 \\ 0 & -2 & 3 & -1 \end{array} \right).$$

We use Gauss-Jordan elimination to bring the matrix in reduced row echelon form:

$$\begin{aligned} \left(\begin{array}{ccc|c} -2 & 5 & 3 & 5 \\ 0 & -2 & 3 & -1 \end{array} \right) & \begin{matrix} R_1 \rightarrow -R_1/2 \\ R_2 \rightarrow -R_2/2 \end{matrix} \rightarrow \left(\begin{array}{ccc|c} 1 & -5/2 & -3/2 & -5/2 \\ 0 & 1 & -3/2 & 1/2 \end{array} \right) \begin{matrix} R_1 \rightarrow 5/2 R_2 + R_1 \\ \\ \end{matrix} \\ & \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -21/4 & -5/4 \\ 0 & 1 & -3/2 & 1/2 \end{array} \right) \begin{matrix} R_1 \rightarrow 5/2 R_2 + R_1 \\ \\ \end{matrix}. \end{aligned}$$

Hence a solution (x, y, z) satisfies

$$\begin{cases} x - \frac{21z}{4} = -\frac{5}{4} \\ y - \frac{3z}{2} = \frac{1}{2} \end{cases}.$$

We set $z = t$ and write

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 21t/4 - 5/4 \\ 3t/2 + 1/2 \\ t \end{pmatrix} = \begin{pmatrix} -5/4 \\ 1/2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 21/4 \\ 3/2 \\ 1 \end{pmatrix}.$$

Hence the intersection of the two planes is a line through $(-5/4, 1/2, 0)$ along the vector $\langle 21/4, 3/2, 1 \rangle$.

12. Use Gauss Jordan elimination to solve the system

$$x + 2y - z = 3$$

$$5x + 4y + z = 3$$

$$2x + y = -1$$

$$2x + 2y - z = 1$$

Solution: The augmented matrix of the system

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 5 & 4 & 1 & 3 \\ 2 & 1 & 0 & -1 \\ 2 & 2 & -1 & 1 \end{array} \right).$$

The steps in the Gauss-Jordan elimination are

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 5 & 4 & 1 & 3 \\ 2 & 1 & 0 & -1 \\ 2 & 2 & -1 & 1 \end{array} \right) & \begin{array}{l} R_2 \rightarrow R_2 - 5R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 6 & -12 \\ 0 & -3 & 2 & -7 \\ 0 & -2 & 1 & -5 \end{array} \right) \begin{array}{l} R_2 \rightarrow -R_2/6 \\ \\ \\ \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & -3 & 2 & -7 \\ 0 & -2 & 1 & -5 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - 2R_2 \\ R_3 \rightarrow 3R_2 + R_3 \\ R_4 \rightarrow R_4 + 2R_2 \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -1 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - R_3 \\ R_4 \rightarrow R_4 - R_3 \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} R_3 \rightarrow -R_3 \\ \\ \\ \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right). \end{aligned}$$

Hence the unique solution of the system is $(-2, 3, 1)$.

13. Find basic solutions of the homogeneous system

$$x_1 - x_2 + 3x_3 - x_4 = 0$$

$$x_1 + x_2 + x_3 + 3x_4 = 0$$

$$2x_1 + x_2 + 3x_3 + 4x_4 = 0$$

$$3x_1 + x_2 + 5x_3 + 5x_4 = 0.$$

Solution: The augmented matrix of the system is

$$\left(\begin{array}{cccc|c} 1 & -1 & 3 & -1 & 0 \\ 1 & 1 & 1 & 3 & 0 \\ 2 & 1 & 3 & 4 & 0 \\ 3 & 1 & 5 & 5 & 0 \end{array}\right).$$

We bring the matrix in reduced row echelon form by the following sequence of row operations

$$\begin{aligned} \left(\begin{array}{cccc|c} 1 & -1 & 3 & -1 & 0 \\ 1 & 1 & 1 & 3 & 0 \\ 2 & 1 & 3 & 4 & 0 \\ 3 & 1 & 5 & 5 & 0 \end{array}\right) & \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array} \rightarrow \left(\begin{array}{cccc|c} 1 & -1 & 3 & -1 & 0 \\ 0 & 2 & -2 & 4 & 0 \\ 0 & 3 & -3 & 6 & 0 \\ 0 & 4 & -4 & 8 & 0 \end{array}\right) \begin{array}{l} R_2 \rightarrow R_2/2 \\ R_3 \rightarrow R_3/3 \\ R_4 \rightarrow R_4/4 \end{array} \\ & \rightarrow \left(\begin{array}{cccc|c} 1 & -1 & 3 & -1 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 2 & 0 \end{array}\right) \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array} \\ & \rightarrow \left(\begin{array}{cccc|c} 1 & -1 & 3 & -1 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \begin{array}{l} R_1 \rightarrow R_1 + R_2 \end{array} \\ & \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right). \end{aligned}$$

Translating back into equations gives

$$\begin{aligned} x_1 + 2x_3 + x_4 &= 0 \\ x_2 - x_3 + 2x_4 &= 0, \end{aligned}$$

which is equivalent to

$$\begin{aligned} x_1 &= -2x_3 - x_4 \\ x_2 &= x_3 - 2x_4. \end{aligned}$$

If we define the parameters $t = x_3$ and $s = x_4$ we find that the general solution of the system is

$$\begin{aligned} (x_1, x_2, x_3, x_4) &= (-2t - s, t - 2s, t, s) \\ &= t(-2, 1, 1, 0) + s(-1, -2, 0, 1). \end{aligned}$$

Hence we have two basic solutions

$$(-2, 1, 1, 0), \quad (-1, -2, 0, 1).$$

14. Consider the system

$$\begin{aligned} -x_1 - x_2 + 2x_3 - 4x_4 &= 0 \\ 2x_1 + x_2 - 4x_3 + 5x_4 &= 0 \\ 4x_1 + x_2 - 8x_3 + 7x_4 &= 0. \end{aligned}$$

(a) Without solving the system, show that it cannot have a unique solution.

Solution: The system has $n = 4$ unknowns and $m = 3$ equations. The rank r of the system is less than or equal to 3. We know that a system can only have a unique solution when $r = n$, which is impossible here since $r \leq 3 < 4 = n$.

- (b) Use Gauss-Jordan elimination to solve the system.

Solution: The augmented matrix is

$$\left(\begin{array}{cccc|c} -1 & -1 & 2 & -4 & 0 \\ 2 & 1 & -4 & 5 & 0 \\ 4 & 1 & -8 & 7 & 0 \end{array} \right).$$

We obtain the reduced row echelon form by performing the following sequence of elementary row operations:

$$\begin{aligned} \left(\begin{array}{cccc|c} -1 & -1 & 2 & -4 & 0 \\ 2 & 1 & -4 & 5 & 0 \\ 4 & 1 & -8 & 7 & 0 \end{array} \right) & \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 + 4R_1 \end{array} \longrightarrow \left(\begin{array}{cccc|c} -1 & -1 & 2 & -4 & 0 \\ 0 & -1 & 0 & -3 & 0 \\ 0 & -3 & 0 & -9 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow -R_1 \\ R_2 \rightarrow -R_2 \\ R_3 \rightarrow -R_3/3 \end{array} \\ & \longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & -2 & 4 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 3 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow -R_1 \\ R_2 \rightarrow -R_2 \\ R_3 \rightarrow -R_3/3 \end{array} \\ & \longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & -2 & 4 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 3 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - R_2 \\ R_3 \rightarrow R_3 - R_2 \end{array} \\ & \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right). \end{aligned}$$

The reduced row echelon form tells us that the solution is given by all (x_1, x_2, x_3, x_4) such that

$$\begin{aligned} x_1 - 2x_3 + x_4 &= 0 \\ x_2 + 3x_4 &= 0, \end{aligned}$$

or equivalently

$$\begin{aligned} x_1 &= 2x_3 - x_4 \\ x_2 &= -3x_4. \end{aligned}$$

We set $s = x_3$ and $t = x_4$ and find that the general solution is

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2s - t \\ -3t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

- (c) Find basic solutions of the system.

Solution: The solutions $\begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}$ are basic solutions, since every solution of the system

can be written as a linear combination of these two solutions.